

A FULL ENGINE-RACK FLEET — WEEKS AHEAD OF THE LINE.

When a Tier-1 supplier needed returnable engine racks faster than their usual source could deliver, SPF designed, welded, and finished the fleet **under one roof** — and hit the launch.

~6-8

WEEKS TO DELIVER
(~HALF THE QUOTED TIME)

300+

RETURNABLE ENGINE
RACKS DELIVERED

ON-TIME

TO THE PROGRAM
RAMP / LAUNCH

THE CHALLENGE

A Tier-1 automotive supplier needed returnable steel racks to move finished engines between its machining and assembly operations — and needed them **fast**, ahead of a program ramp. Their existing rack source quoted roughly twice that — around **[12–16 weeks]** — a lead time that wouldn't make the launch date and put the line start at risk.

WHAT SPF DID

SPF engineered the engine rack in-house with SolidWorks, then cut, robotic-welded, and powder-coated the full fleet in one Byron facility — no brokered steel and no outside coater's queue to wait in. One purchase order, one accountable team, and short Southeast freight kept the whole job moving on a single clock.

THE RESULT

The full 300+-rack fleet was delivered in about **6–8 weeks** — roughly half the lead time quoted elsewhere — in time for the program ramp. The returnable racks are built to run **[N]** trips, protecting finished engines in transit and replacing **[expendable packaging]** trip after trip.



RETURNABLE STEEL RACKS · POWDER-COATED

“[SPF turned our engine racks around in about half the time we expected — and they hit our launch.]”

[MATERIALS MANAGER, TIER-1 AUTOMOTIVE SUPPLIER]

WHY SPF

FAST TURNAROUND

ONE ROOF

ROBOTIC WELDING

RETURNABLE

INDUSTRIES

AUTOMOTIVE TIER-1

POWERTRAIN

Need racks on a deadline? sales@southernperfection.com · (478) 956-4442 · toll-free (800) 237-4726

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A RACK ENGINEERED FOR A PART NOTHING ELSE FIT.

When an off-highway OEM had an oversized, awkward weldment with no standard rack to hold it, SPF's **in-house engineering** turned a rough idea into a buildable, ergonomic returnable rack.

1

DESIGN ROUND –
FIRST-PASS APPROVAL

1-3K LB

RATED LOAD
PER RACK

5-10

PARTS STAGED
PER RACK

THE CHALLENGE

An off-highway / heavy-equipment OEM needed to move large weldments — **[frames / axles / booms]** — between weld, machining, and assembly. The parts were heavy and awkward, no off-the-shelf rack fit, and manual handling raised damage and ergonomic concerns.

WHAT SPF DID

SPF's SolidWorks team took the part data and a rough concept and engineered a returnable rack around it — load paths, fork access, and operator ergonomics designed in. A prototype was reviewed, adjusted, and moved to production, all in-house.

THE RESULT

Approved on the **first design pass** and rated to 1,000–3,000 lb, the rack stages 5–10 parts safely, cut **[manual handling]**, and became the standard design for the program's follow-on parts.



CUSTOM-ENGINEERED RETURNABLE RACK · FOLDING STANCHIONS, MESH DECK

“[The rack did what three vendors said couldn't be done for that part.]”

[MANUFACTURING ENGINEER, HEAVY-EQUIPMENT OEM]

WHY SPF

BUILD-TO-PRINT

IN-HOUSE DESIGN

ROBOTIC WELDING

ONE ROOF

INDUSTRIES

HEAVY EQUIPMENT

INDUSTRIAL MACHINERY

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AN AGING FLEET, BACK TO SPEC FOR A FRACTION OF NEW.

Rather than replace a worn fleet of returnable racks, a manufacturer had SPF **inspect, repair, and re-finish** them — back in service in days, at a fraction of new-build cost.

~40% OF NEW-BUILD COST	75-200 RACKS REFURBISHED & RETURNED	1-2 WKS TURNAROUND PER BATCH
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THE CHALLENGE

After **[several years]** in a returnable loop, a manufacturer's fleet of 75–200 steel racks was bent, cracked, and worn — risking part damage and operator safety. Replacing the fleet outright was costly and slow.

WHAT SPF DID

SPF inspected the fleet, re-welded and repaired damaged members, replaced what was beyond saving, and re-finished every rack back to original spec. Because SPF builds racks like these, the failure points were already familiar.

THE RESULT

75–200 racks returned to service at about **40% of new-build cost** in 1–2 weeks per batch — extending fleet life by **[several years]** and restoring safe, damage-free handling.

“[They turned our beat-up racks around faster and cheaper than buying new.]”

[PLANT MATERIALS MANAGER]

WHY SPF

- REPAIR & REFURB
- FAST TURNAROUND
- FLEET ECONOMICS
- ONE ROOF

INDUSTRIES

- AUTOMOTIVE
- GENERAL MFG

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1,000+ IDENTICAL RACKS, ON TIME FOR LAUNCH.

An OEM ramping a new line needed a large fleet of identical returnable racks on a hard deadline — SPF's **robotic welding** delivered the volume and the weld consistency.

1,000+

RETURNABLE RACKS
DELIVERED

100%

ON-TIME TO THE
LAUNCH DATE

ROBOTIC

WELDED FOR RACK-TO-
RACK CONSISTENCY

THE CHALLENGE

An automotive OEM ramping a new **[line / model]** needed more than 1,000 identical returnable racks by a fixed launch date. Volume was high, the timeline tight, and weld consistency across every rack was critical.

WHAT SPF DID

SPF ran the weldments through FANUC robotic welding for repeatable, spec-consistent joints at volume, scaling output to hold the launch window. Cutting, welding, and powder coat all stayed in-house on one PO.

THE RESULT

Over **1,000 racks delivered 100% on time** for launch — every weldment robotic-welded for rack-to-rack consistency — and the account became a repeat program for follow-on parts.



A STAGED FLEET OF IDENTICAL RACKS · POWDER-COATED

“[Every rack was identical and they all showed up on schedule.]”

[SUPPLIER QUALITY, AUTOMOTIVE OEM]

WHY SPF

ROBOTIC WELDING

PRODUCTION CAPACITY

ON-TIME

ONE ROOF

INDUSTRIES

AUTOMOTIVE OEM

EV / BATTERY

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THREE VENDORS, ONE LEAD TIME — UNDER ONE ROOF.

A plant juggling separate cut, weld, and coat suppliers moved the whole job to SPF — **fewer handoffs, shorter lead times, one accountable team.**

~50%

LEAD TIME
CUT

3 → 1

SUPPLIERS
CONSOLIDATED

20-30%

LOWER FREIGHT
COST

THE CHALLENGE

A manufacturer was sourcing racks across three vendors — one cut, one welded, one coated. Handoffs stretched lead times, quality issues bounced between suppliers, and out-of-region freight added cost and delay.

WHAT SPF DID

SPF took the entire job in-house — cut, form, weld, and finish on one floor in Byron, on a single purchase order. Short Southeast freight replaced long hauls, and one team owned the result end to end.

THE RESULT

Lead time was **cut roughly in half**, supplier count went from 3 to 1, and freight fell 20–30% with regional sourcing — all on one PO, with a shop the customer can visit and audit.



ONE FACILITY IN BYRON — CUT, WELD & FINISH
UNDER ONE ROOF

“[One PO, one number to call, and the
racks come in faster.]”

[PROCUREMENT LEAD, MANUFACTURER]

WHY SPF

ONE ROOF

ONE PO

SE FREIGHT

ISO 9001

INDUSTRIES

GENERAL MFG

INDUSTRIAL MACHINERY

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NON-FLIGHT TOOLING & GSE, BUILT TO A QUALITY SPEC.

A regulated aerospace / defense shop needed durable, traceable tooling carts and ground-support equipment — SPF built to print under **ISO 9001** and **CAGE 2W654**.

100%

SPEC CONFORMANCE
/ ACCEPTANCE

25-75

CARTS & STANDS
DELIVERED

ONGOING

AUDITED, REPEAT
SUPPLIER

THE CHALLENGE

An aerospace & defense manufacturer needed non-flight tooling carts, work stands, and transport GSE — precise, durable, and built to a documented quality standard with full traceability.

WHAT SPF DID

SPF machined, robotic-welded, and finished the equipment to print under its ISO 9001 quality system and CAGE 2W654 registration — one accountable shop for fabrication, finishing, and documentation.

THE RESULT

25–75 carts and stands delivered at **100% conformance**, with the documentation trail the program required — and SPF onboarded as an audited, **ongoing supplier**.



MATERIAL-HANDLING TRANSPORT CART

“[They hit our spec and our paperwork requirements without hand-holding.]”

[PROCUREMENT, AEROSPACE & DEFENSE PROGRAM]

WHY SPF

ISO 9001

CAGE 2W654

BUILD-TO-PRINT

ONE ROOF

INDUSTRIES

AEROSPACE & DEFENSE

INDUSTRIAL MACHINERY

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